Kaili (Kelly) Zhang

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RESEARCH INTERESTS

- Climate justice
- Ecological Footprint
- Urban Analytics

EDUCATION & HONORS

Texas A&M University **Aug. 2024-Present**

- Ph.D. Student | **Major**: Urban and Regional Science| **GPA**: 3.84/4.00
- **Selected Courses**: Spatial Statistics, Deep Learning, SPTP: Generative AI, Machine Learning, SPTP: Graph Machine Learning, Climate Adaptation & Hazard Mitigation

Wuhan University **Sep. 2021-Jun. 2024**

- **Degree**: Master | **Major**: Urban and Rural Planning | **GPA**: 3.67/4.00
- **Scholarship**: Scholarship for Outstanding Students, Second Prize, 2023

Wuhan University **Sep. 2015-Jun. 2020**

- **Degree**: Bachelor | **Major**: Urban and Rural Planning | **GPA**: 3.42/4.00
- **Scholarship**: UCGD Inspirational Scholarship (first prize level), 2019; "Xianzi Zeng" Scholarship (first prize level), 2016, 2017&2018

PUBLICATIONS

[1] Xu, B., **Zhang, K**., Li, L., Yu, Y., Kaitlin, K., Goldstein, B., Dadashova B., Newman G., Wu, L., Liu, Z., Ye, X., Lin, D., & Jiao, W. Higher Ecological Footprints in Cities Driven by Service and Income Effect. 2026. *Nature Communications.* (Accepted). DOI: 10.1038/s41467-026-78515-8

- Data Collection, Formal analysis

[2] Zhang, Y., **Zhang, K***., Ding, Y., Wu, C., & Li, D. (2026). Subtropical Climate and Intraday Temperature Variability: Mental Health Risks and Modifying Roles of Demographic and Seasonal Factors in Texas. *Sustainable Cities and Society*, 149, 107751. DOI: 10.1016/j.scs.2026.107751

- Conceptualization, Methodology, Data Collection, Formal analysis, Visualization, Writing - Original Draft, Writing - Review & Editing

[3] Yang, Y., Gong, W., **Zhang, K**., Li, H., Zhou, B., Li, Z., Ye, X., & Zou, L. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. arXiv:2606.21819. *Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26), November 3–6, 2026.*

- Data Collection, Formal analysis, Visualization, Writing - Original Draft

[4] Li, X., **Zhang, K**., Shi, W., & Li, Z. (2026). Multi-objective optimization of urban land use across multiple scales: Balancing ecosystem services supply and demand with an integrated NSGA-II–PLUS model. *Environmental Impact Assessment Review*, 121, 108531. DOI: 10.1016/j.eiar.2026.108531.

- Data Collection, Formal analysis

[5] **Zhang, K**., Xu, B. Cities Intensify Income-Driven Ecological Footprint Growth. (Under Review)

- Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing

[6] **Zhang, K**., & Gong, W. The uneven geography of post-fire recovery: An LLM-assisted assessment of rebuilding after six California wildfires. (Working paper)

- Conceptualization, Methodology, Data Collection, Formal analysis, Visualization, Writing - Original Draft, Writing - Review & Editing

[7] **Zhang, K**., Zhang, Y., Li, D., & Xu, B. Monthly water stress and drought-related water redistribution from U.S. data centers. (Working paper)

- Conceptualization, Methodology, Data Collection, Formal analysis, Visualization, Writing - Original Draft, Writing - Review & Editing

[8] Li, Z., **Zhang, K**., Liu, X., & Liu, D. (2026). Impacts and threshold effects of urban human settlements upon

job-housing balance in megacities: A case study of Wuhan metropolitan development area. *Journal of Natural Resources*, 41(2), 500–515. DOI: 10.31497/zrzyxb.20260211. (In Chinese)

- Conceptualization, Methodology, Data Collection, Formal analysis, Visualization, Writing - Original Draft, Writing - Review & Editing

CONFERENCE PRESENTATIONS

[1] Zhang, K., Xu, B. Climate Justice Indicators and Actions: A Systematic Literature Review. *2026 Annual Meeting of American Association of Geographers*, March 17-21, 2026, San Francisco, California, USA (Oral).

[2] Zhang, K., Xu, B. Higher Neighborhood Ecological Footprint in villages and towns than cities and suburbs in the United States. *2025 Annual Meeting of American Association of Geographers*, March 24-28, 2025, Detroit, Michigan, USA (Oral).

RESEARCH EXPERIENCE

DisasterVLP: Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models

GitHub: https://github.com/kellyzhang8/DisasterVLP **Agu. 2025-Feb. 2026**

- Designed a multi-agent GeoAI framework integrating multimodal data (satellite, street view, text, temporal signals) for hyperlocal disaster damage assessment.
- Leveraged vision–language models (e.g., GPT, Gemini) for zero/few-shot analysis, cross-view reasoning, and automated disaster intelligence reporting.

Mississippi Band of Choctaw Indians ("MBCI") Forest Carbon Project Case Study Dec. 2025-Feb. 2026

- Built an Alpha Earth embedding–driven Machine learning mapping pipeline to produce annual 10-m land-cover and forest-height products (2017–2024), anchored to GLC_FCS10 (2023) and global CHM canopy height (2020).

A Research on Implementation-oriented Eco-friendly Low-carbon Community [Grant number: 250071521] **Sep. 2022-Nov. 2022**

- Summarized the literature on the measurement of mixed land use, constructed an indicator system, and used Python to batch carry out mixed land use measurement for low-carbon communities in Wuhan City.
- Proposed strategies for low-carbon community construction in Wuhan, with an in-depth empirical exploration of cases of low-carbon communities and carbon reduction strategies.

EXTRACURRICULAR ACTIVITIES

" Chutian Cup" City Model Skills Competition in Hubei Province **Dec. 2022-May. 2023**

- Won the **1st** prize; Pioneered the idea of measuring "urban shadow zones" in the Wuhan metropolitan area, and constructed an index system that integrates dynamic relationships between communities and static land use.

The 2nd National College Student Land Space Planning And Design Competition **Jun. 2022-Sep. 2022**

- Achieved **2nd** prize and "Best Analysis" for the competition; Analyzed how the accessibility of public services affected residents' happiness with questionnaire data.

KEY STRENGTHS

- Expertise in **machine learning, deep learning, and generative AI for geospatial and urban analysis**
- Strong **data processing and visualization** skills across multi-source spatial and survey datasets
- **Technical Proficiency**, including Python, R, Google Earth Engine, Stata, ArcGIS, PS, AI, ID, CAD
- Effective **teamwork, project coordination, and cross-disciplinary collaboration**
- **Language Proficiency**: English (fluent), Mandarin (First Language)